

KMAT Truck Cost Model

Final Project Documentation and Demo Guide

Snowflake cost model with governed machine learning

Current state: Completed in governed SHADOW mode

The official cost is still deterministic. ML models give advice only. Runtime model authority is installed but disabled.

1. Project overview

In one line: The system reads a truck configuration or RFQ, calculates cost, checks risk, and keeps every ML decision controlled and auditable.

Main problem: Truck cost changes with engine, cab, wheel, colour, material price, scrap, labour, tooling and overhead.

Main solution: A Snowflake-based model calculates the official cost and uses ML only as a safe advisory layer.

2. How the system works

1. Manual configuration or RFQ document

v

2. Configuration extraction and validation

v

3. BMCS trust and review check

v

4. Deterministic truck cost calculation

v

5. CSS, FMIS and TDS advisory signals

v

6. Governed fallback and safety checks

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7. Cost output, audit and monitoring

The final output includes configured cost, risk-adjusted cost, price range, trust status, model lineage and audit history.

3. ML models used

Model	Simple purpose	Current authority
BMCS	Checks configuration mapping confidence.	Review only. Never changes cost.
CSS	Estimates scrap risk.	ML is advisory. Rule value stays official.
FMIS	Estimates material market multiplier.	ML is advisory. Rule value stays official.
TDS	Estimates tooling difficulty factor.	ML is advisory. Rule value stays official.

4. Safety and governance

- Official cost authority is deterministic.
- All four models are in SHADOW mode.
- Runtime model authority is disabled.
- Each domain has a kill switch and deterministic fallback.
- The application uses secure procedures and separate Snowflake roles.
- Important actions are written to audit tables.
- BMCS never changes cost directly.

Fallback order: approved engineer override -> governed ML when allowed -> deterministic rule -> safe default.

This design prevents an unsafe model value from silently changing the official truck cost.

5. Simple architecture

Layer	Purpose
CORE_INPUT	RFQ, configuration, parameters and production context
CORE_INTERNAL	Deterministic cost and controlled RFQ procedures
CORE_ML	Models, policies, decisions, monitoring, runtime controls and audits
CORE_OUTPUT	Cost summary, component cost, operation cost and overhead cost
Streamlit	User input, results, governance status and monitoring screens

6. Main procedures

RUN_KMAT_COST_SIMULATION_GOVERNED_V3 - Runs the normal governed truck cost flow.

ACTIVATE_APPROVED_ML_DEPLOYMENT_SECURE_V1 - Future controlled activation entry point. Not used in the current SHADOW state.

RESOLVE_ML_RUNTIME_AUTHORITY_SECURE_V1 - Checks whether a candidate value may be used or must fall back.

SUBMIT / REVIEW_ML_DECISION_OVERRIDE_V2 - Two-person controlled override workflow.

Normal Streamlit simulations still use the governed V3 cost procedure. Activation is a separate administrative action.

7. Final project status

Area	Status
Deterministic cost engine	Complete
CSS, FMIS, TDS and BMCS models	Complete
Governed fallback hierarchy	Complete
Monitoring and feedback	Complete
Backtesting and pilot controls	Complete
Runtime safety and kill switches	Complete
Privilege hardening	Complete
Secure Streamlit integration	Complete
Final regression validation	Passed

8. Final verified state

Active policies	4
SHADOW policies	4
Enabled runtime authorities	0
Runtime decisions	0
Security status	COMPLIANT
Bypass violations	0
Missing grants	0
Official cost authority	DETERMINISTIC

Project completion means the governed SHADOW system is ready and tested. It does not mean ML is already active in production.

9. Short demo script (about 6 minutes)

1. Open the Streamlit home page and explain the goal: accurate truck cost with safe ML governance.
2. Show a manual truck configuration or upload one RFQ document.
3. Run the simulation and show the configured cost, risk-adjusted cost and target price.
4. Open the governed decision section and show CSS, FMIS, TDS and BMCS values.
5. Explain that rule values are official and ML values are advisory.
6. Show Security State = COMPLIANT, Runtime Paths Open = 0 and Enabled Authorities = 0.
7. Open monitoring or audit history and show that decisions are traceable.
8. Finish by saying the system is ready for a future controlled pilot, but it is safely blocked today.

10. What to say during the demo

"This project is a governed truck cost model built in Snowflake. The deterministic engine calculates the official cost. CSS, FMIS, TDS and BMCS add advisory intelligence. The system includes fallback rules, monitoring, approval gates, kill switches, secure roles and audit trails. The final verified state is SHADOW mode, so ML cannot change official cost without a future approved activation."

11. Important demo rules

- Do not activate a model during the demo.
- Do not enable runtime authority.
- Do not run rollback or controlled-operation example scripts.
- Use test RFQs and test simulation IDs only.

12. Simple interview explanation

30-second answer: I built a Snowflake-based truck cost model that calculates configured cost and price. I added four ML models for mapping confidence, scrap, material market movement and tooling difficulty. The ML layer is governed with policies, fallback logic, monitoring, pilot gates, runtime controls, secure roles and audit trails. The current system is fully tested in SHADOW mode, so deterministic logic remains the official cost authority.

13. Key technical points

- Snowflake SQL, Snowpark and Streamlit
- Deterministic cost calculation with versioned rules
- XGBoost and calibrated classification/regression models
- Cortex-based RFQ extraction and explanation support
- Governed model lifecycle from shadow testing to controlled activation
- Role-based security, secure procedures and audit logging
- Fallback, kill-switch and monitoring design

14. Honest final claim

The project is complete as a governed SHADOW deployment. The ML models are not active production cost authorities. A real controlled pilot would need approved business data, assigned users, operational ownership and formal sign-off.